XING GUANGJIE

jiengup@gmail.com | (+86) 13995697434 | Personal Webpage | github.com/jiengup

EDUCATION

Huazhong University of Science and Technology (HUST) **Wuhan, China**
Master of Engineering in Computer Architecture 2022/09 – 2025/06

- **Courses**: Distributed Systems, Deep Learning Systems, Database Systems, etc.
- **Research Area**: Block Storage System / Cloud Resource Scheduling
- **Achievement**: GPA 88.4/100, National Scholarship 2024 (Top 0.2% national-wide), Academic Scholarships× 2
- **Serve As**: Teaching Assistant, Research Assistant, Leader of Data Intelligence Group

Huazhong Agricultural University **Wuhan, China**
Bachelor of Engineering in Computer Science and Technology 2018/09 – 2022/06

- **Courses**: Operating Systems, Computer Networks, Computer Architecture, Machine Learning, etc.
- **Achievement**: GPA: 3.9/4.0 (2^{ed}/136), Top 10 Students of HZAU, National Scholarship 2018, 2020
- **Experiences**: ACM-ICPC, Volunteer Teaching in Underdevelopment, Youth Volunteer Department

PUBLICATIONS

MINOS: Towards a Theoretical WAF = 1 in Log-Structured Storage
Under Review at 24th USENIX Conference on File and Storage Technologies (FAST 2026) Second Author

DuoAdmit: Dual-Layer Cache Admission for Load-Balancing Hybrid-Redundancy Block Storage
Accepted by ACM Symposium on Cloud Computing 202 (SoCC 2025) Second Author

VDMig: An Adaptive Virtual Disk Migration Scheme for Cloud Block Storage System
Published on IEEE 42nd International Conference on Computer Design (ICCD 2024) First Author

FLOWS: Balanced MRC Profiling for Heterogeneous Object-Size Cache
Published on Nineteenth European Conference on Computer Systems (EurySys 2024) Core Author

WORK EXPERIENCE

Baidu AI Cloud **Shenzhen, China**
Senior Software R&D, ACG CFS/PFS team (Top 1% Campus Recruitment) 2025/06 – Present

- **POSIX-Compatible CFS Optimization**: Enhance performance for flat structure metadata operations (checkpoint readdir, 1PC setattr, etc.) for industry-leading large-scale distributed cloud file storage based on NFS 4.
- **Building PFS for AI infra**: Develop features of Lifecycle Management of parallel file systems (data flow and hierarchy between PFS and object storage) and high-performance stand-alone storage engine.

Tencent Cloud **Shenzhen, China**
Software Engineer Intern, TEG CBS team 2024/05 – 2024/08

- **Storage Engine Dev**: Decouple SPDK FTL module to support the append-only engine for custom Write Separation and Garbage Collection strategies on general NVMe SSDs. The implementation of the state-of-art WS and GC algorithm tests show that the WAF can be reduced from 4.31 to 1.59 under zipfan:0.8 simulation and from 2.13 to 1.56 under online trace.

Tencent Cloud **Shenzhen, China**
Storage System R&D Intern, TEG CBS team 2022/01 – 2022/05

- **Intelligent Resource Scheduling**: Characterize online workload's access patterns and performs time-series prediction, achieves an increase of up to 20% in utilization rate across multiple resource dimensions.
- **Prioritized Flow Control**: Proposed and implemented a flow control prototype based on a lock-free queue.

Sensetime Inc. **Shanghai, China**
CV Research Intern 2021/07 – 2021/11

- **ICCV-MFR21**: Performed mixup augmentation on over a million image dataset, achieved 1st in the MS1M Track.
- **Embedded Deployment**: Reproduce the small architecture model, perform knowledge distillation, complete quantization, and conduct real-time deployment testing on embedded devices.

RESEARCH PROJECTS

Optimizing Data Separation for Multi-Stream Log-Structured Storage Systems
github.com/jiengup/spdk/tree/master/lib/ftl 2024/12 – 2025/08

- **OP-Minimized Oracle** finds the optimal separation thresholds to achieve 1.0 WAF minimal OP space
- **Two-Stage Seperation** classifies incoming data into virtual streams based on prediction, which are then mapped to limited system-level physical streams via an adaptive clustering process
- **Evaluations** on real-world workloads show saving of GC writes by 37.4% and improvement of throughput by 14.2%

Cache Admission Framework for Hybrid-Redundancy Block Storage 2024/07 – 2025/05

- **Model and Tune** optimal cache-layer admission ratio for each node of HRBS using Linear Programming
- **Value-Based Admission policy** reduces traffic amplification and cuts EC-layer flush burst during node failure
- **HRBS Prototype** improves cluster throughput by 10.6% under single-node failures and 51.2% under multi-node failures

An Adaptive Virtual Disk Migration Scheme For Cloud Block Storage System 2023/12 – 2024/05

- **Classify Cloud Virtual Disks** characterized by short bursts workload based on user subscription information
- **Trigger and Timer Migrator** are proposed to distribute and schedule CVDs with different access patterns
- **Experiments** shows that the schema can reduce the average load imbalance between cluster machines by 82.4%

Accurent and Balanced MRC Profiling for Heterogeneous Object-Size Cache
github.com/jiengup/cache-allocator 2023/07 – 2024/02

- **Weighted Sampling** implemented in libCachesim achieves more accurate Byte-MRC estimation in CDN cache.
- **Multi-Tenant Allocation** is modeled as a multi-objective optimization and solved by the proposed DP algorithm
- **Simulations** indicate that proposed method can reduce the overall miss ratio by 23.4% compared to the state-of-the-art

PATENTS

用户请求处理方法、装置、设备和介质
Invention Patent CN115242727B 2023/08

AWARDS

The 45th International Collegiate Programming Contest Asia Regional **Bronze Medal** 2021/03

2020 China College Programming Contest **Bronze Medal** 2020/10

The 8th China International Internet+ Innovation and Entrepreneurship Competition **Gold Medal** 2023/04

SKILLS & INTERESTS

- **Coding**: Python, C/C++, Go, Rust, CCF-CSP 310′(Top 1.04% national-wide)
- **Interests**: Fitness, Photography, Large Language Models